

Introduction to Optimization Project 1

Jun Moon

junmoon@hanyang.ac.kr

Project 1

- Individual Project
- Submission with zip file (압축해서 제출해주세요)
 - Programming files (py-files)
 - Solution PDF
 - PPT Slides
 - Presentation MP4 (25 mins)

Submission

- 1. PPT Slides of your presentation**
- 2. The explanation of your solution in English (PDF Format)**
- 3. Python Codes**
- 4. Presentation MP4 (25 mins)**

Submission 예시

Name	
	그룹_1_답안.pdf
	그룹_1_발표자료.pptx
> 	그룹_1_조원_1_홍길동_프로그램파일
> 	그룹_1_조원_2_홍길동_프로그램파일
> 	그룹_1_조원_3_홍길동_프로그램파일
> 	그룹_1_조원_4_홍길동_프로그램파일

그룹_1_발표동영상.mp4

제출물 파일들 예시

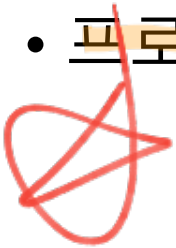
- 모든 조원들의 결과가 있어야함
- 가장 좋은 결과를 선정하고 논리적인 근거 제시

Note

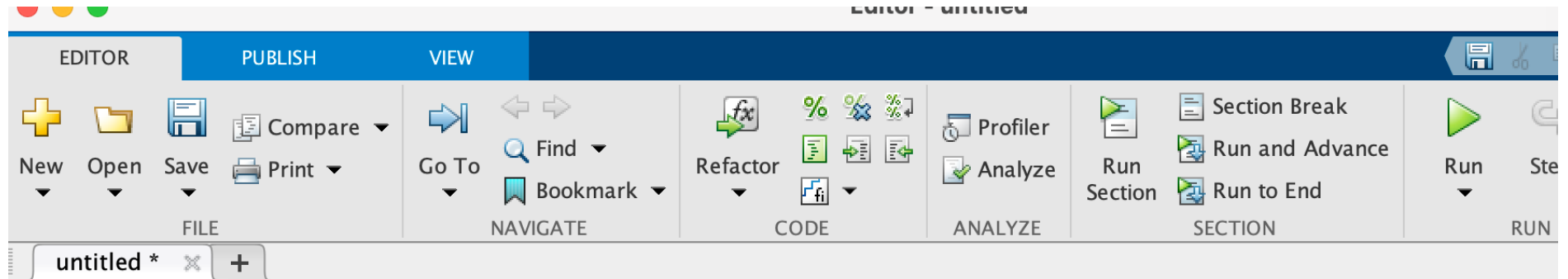
- **You can use Python to solve the problem**
- **You have to construct the algorithm by yourself**
- **You are not able to use any Python optimization toolbox**
 - **You need to implement the numerical optimization algorithms**

공지사항

- 프로그래밍은 모두 참여해야 할 것 입니다
- 조원들 프로그래밍 중 가장 좋은 결과를 선택하셔야 합니다
- 프로그래밍 매 라인마다 주석으로 설명이 있어야 합니다.



공지사항

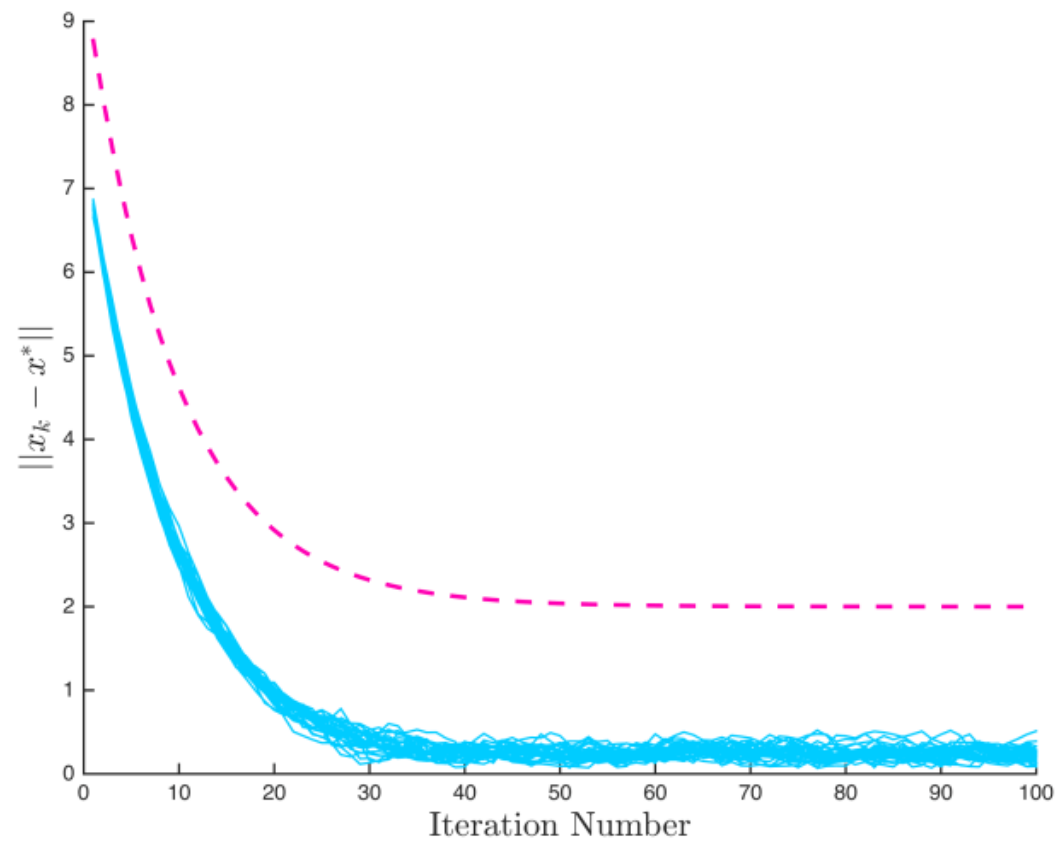


Required Python Libraries

- Numpy
- Matplotlib
- Scipy
- Pandas
- etc

Example of Convergence plot

- y axis : error (norm square)
- x axis: iteration number



Problem 1

$$f(x_1, x_2) = 1 + 2x_1 + 3(x_1^2 + x_2^2) + 4x_1x_2$$

$$f(x_1, x_2) = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^\top \begin{bmatrix} 3 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 16 & 23 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \pi^2$$

$$f(x, y) = 3(x^2 + y^2) + 4xy + 5x + 6y + 7$$

- ✓ 1. Show that the functions are convex.
- ✓ 2. Obtain the optimal solution to the above problems using CVX
- ✓ 3. Obtain the optimal solution and optimal cost for each function by your hand
4. Use Python to design the gradient descent algorithm and find the optimal solution and optimal value. Also, obtain the convergence plot
5. Use Python to design the steepest (optimal step size) gradient descent and find the optimal solution and optimal value. Also, obtain the convergence plot
6. Let x^* be the optimal solution obtained by 2. Let $x_{Gradient}^*$ be the optimal solution obtained by 2. Let $x_{SteepestGradient}^*$ be the optimal solution obtained by 2. Plot the convergence plot of $\|x^* - x_{Gradient}^*\|$, $\|x^* - x_{SteepestGradient}^*\|$
7. Discuss the convergence speed in the convergence plots in 5 and 6

Problem 2

Consider the quadratic programming

$$f(x) = \frac{1}{2}x^\top Qx - b^\top x$$

$$\nabla f(x) = Qx - b$$

$$Q = Q^\top > 0, \quad b \in \mathbb{R}^n$$

Use “randn” in numpy to generate a 1000 x 1000 matrix A. Then obtain Q by

$$Q = AA^\top + \rho I$$

Here, rho is any positive number such that Q is positive definite.

Use “randn” in numpy to generate 1000 x 1 vector b

1. Show that the function is convex.
2. Obtain the optimal solution and optimal cost using CVX
3. Use Python to design the gradient descent algorithm and find the optimal solution and optimal value. Also, obtain the convergence plot
4. Use Python to design the steepest (optimal step size) gradient descent and find the optimal solution and optimal value. Also, obtain the convergence plot
5. Use Python to design the Nesterov-2 algorithm and find the optimal solution and optimal value. Also, obtain the convergence plot
6. Let x^* be the optimal solution obtained by 2. Let $x_{Gradient}^*$ be the optimal solution obtained by 3. Let $x_{SteepestGradient}^*$ be the optimal solution obtained by 4. Let $x_{Nesterov-2}^*$ be the optimal solution obtained by the Nesterov-2 in 5. Plot the convergence plot of

$$\|x^* - x_{Gradient}^*\|, \|x^* - x_{SteepestGradient}^*\|, \|x^* - x_{Nesterov-2}^*\|$$
7. Discuss the convergence speed in the convergence plots in 6

Problem 3

$$f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$$

1. Find the optimal solution using the gradient descent method with the initial condition that you choose
2. Try the gradient descent with different initial conditions
 - For 1 and 2, you need to use Python
 - Discuss the convergence speed with different initial conditions
 - Does the convergence depend on the initial condition?

Problem 4

$$\min_{x \in \mathbb{R}^n} c^\top x$$

subject to $Ax \leq b$

$$x_i \geq 0, \quad i = 1, \dots, n$$

note that

$$c \in \mathbb{R}^n, \quad A \in \mathbb{R}^{p \times n} \quad (p \leq n), \quad b \in \mathbb{R}^p$$

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Problem 4

- You have to use either linprog in MATLAB
(<https://kr.mathworks.com/help/optim/ug/linprog.html>)
- Or linprog in Python
(<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html>)

Problem 4

- Use MATLAB or Python to generate random c , A , b data (e.g. for MATLAB)
 - $c = \text{rand}(10,1)$, $A = \text{randn}(8,10)*3+0.2$, $b = -5 + (5+5)*\text{rand}(8,1)$
 - Use `linprog` in MATLAB or Python to find the optimal cost and optimal solution with the random data given above
- Use MATLAB or Python to generate random c , A , b data (e.g. for MATLAB)
 - $c = \text{rand}(100,1)$, $A = \text{randn}(55,100)*5-1.2$, $b = -1 + (1+1)*\text{rand}(55,1)$
 - Use `linprog` in MATLAB or Python to find the optimal cost and optimal solution with the random data given above
- Discuss the location of the optimal solution in terms of the constraints

$$Ax \leq b$$

$$x_i \geq 0, \quad i = 1, \dots, n$$

Problem 5

$$\min_{x \in \mathbb{R}^n} f(x)$$

$$\text{where } f(x) = \|Ax - b\|_Q^2 = (Ax - b)^\top Q(Ax - b)$$

$$A \in \mathbb{R}^{m \times n}$$

$$b \in \mathbb{R}^m$$

$$Q = Q^\top \in \mathbb{R}^{m \times n} : \text{ positive definite matrix}$$

Problem 5

1. Show that the problem, is convex
2. Find the optimal solution to this problem by your hand
3. Let $n=100$, $m=50$, use Python to generate the Gaussian random data of A , b , Q . Then find the optimal solution via the gradient descent method and the solution of 2. Also use CVX to find the optimal solution
4. Let $n=50$, $m=100$, use Python to generate the Gaussian random data of A , b , Q . Then find the optimal solution via the gradient descent method and the solution of 2. Also use CVX to find the optimal solution.
5. Discuss the solutions of 3 and 4. Provide your discussion in terms of the matrix inverse and rank condition